

MS7024

TV Encoder

Register Map

Version: V1.0

Data: 2016-2-24

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Revision History

Date	Version	Author	Comments
2016-2-24	V1.0	MacroSilicon	Basic version

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TOP Registers

Base Address: 0x00

Addr.	Name	7	6	5	4	3	2	1	0
0x04	CORE_CLK_SEL	Reserved			r_2clk_sel	r_vclk_sel	r_dflk_sel	r_vinclk_inv	r_pllsrc_sel
Label		R/W	Description						Default
r_2clk_sel		RW	Tve input v2clk select . 0: same as dvin module clk; 1:from PLLVM clk						0x0
r_vclk_sel		RW	FFo output and tve input vclk select . 0: same as dflicker clk ; 1:half of clk from PLLVM						0x0
r_dflk_sel		RW	deflicker input clk select . 0: same as dvin module clk ; 1: half of dvin module clk						0x0
r_vinclk_inv		RW	Video input clk inv enable. 0: inv disable ; 1: inv enable						0x0
r_pllsrc_sel		RW	PLLVM source clk . 0 : from video input clk ; 1: from OSC clk						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x05	CORE_CLK_EN	Reserved					r_ff_en	r_dflk_en	r_dvin_en
Label		R/W	Description						Default
r_ff_en		RW	Cross_fifo/tve/dac_if module clk enable						0x0
r_dflk_en		RW	Deflicker module clk enable						0x0
r_dvin_en		RW	Dvin module clk enable						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x06	CORE_RST_EN	Reserved				t_tve_rst_b	r_ff_rst_b	r_dflk_rst_b	r_dvin_rstb
Label		R/W	Description						Default
t_tve_rstb		RW	Tve module reset release enable						0x0
r_ff_rst_b		RW	Cross_fifo module reset release enable						0x0
r_dflk_rstb		RW	Deflicker module reset release enable						0x0
r_dvin_rstb		RW	Dvin module reset release enable						0x0

Data Demux and CSC Registers

Base Address: 0x50

Addr.	Name	7	6	5	4	3	2	1	0
0x00	DVIN_SWAP	r_data_swap	r_data_sel			r_data_sync_swap	r_sync_swap		
Label		R/W	Description						Default
r_data_swap		RW	Input video swap : bit[23:0] → bit[0:23]						0x0
r_data_sel		RW	Digital video input data selection,combined with input mode for each kind of input data selection.						0x0
r_data_sync_swap		RW	Input video timing and data swap enable: {dein,hsin,vsin,din[23:0]} → {din[23:0],hsin,vsin,dein}						0x0
r_sync_swap		RW	Input video timing swap : r_sync_swap : 0 1 2 3 4 5 deout: dein dein hsin hsin vsin vsin vsout: vsin hsin vsin dein hsin dein hsout: hsin vsin dein vsin dein hsin						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x01	DVIN_INSEL	r_cut_test	r_crc_tim_sel	r_crc_check_en	r_ccir_cutt_en	r_bit1120_mode	r_bit656_mode	r_8bit_in	r_24bit_in
Label		R/W	Description						Default
r_cut_test		RW	Test to disable blank data						0x0
r_crc_tim_sel		RW	Enable crc timing result						0x0

r_crc_check_en	RW	Crc check enable	0x0
r_ccir_cut_en	RW	Disable blank data enable(disable sav/eav)	0x1
r_bit1120_mode	RW	Input video mode is bit1120	0x0
r_bit656_mode	RW	Input video mode is bit656	0x0
r_8bit_in	RW	8bit video input	0x0
r_24bit_in	RW	24bit video input	0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x02	DVIN_SYN CIN	r_dein_fli p	r_fldin_fli p	r_vsin_fli p	r_hsin_fli p	r_hvshift _en	r_dein_v alid	r_fld_det _en	r_prgs_e n
Label		R/W	Description						Default
r_dein_flip		RW	Input timing de flip						0x0
r_fldin_flip		RW	Input timing field flip						0x0
r_vsin_flip		RW	Input timing v_sync flip						0x0
r_hsin_flip		RW	Input timing h_sync flip						0x0
r_hvshift_en		RW	Pixel shift enable						0x1
r_dein_valid		RW							0x0
r_fld_det_en		RW	Field detection enable						0x1
r_prgs_en		RW	Enable input progressive video						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x03	DVIN_CSC	r_dvalid_ 2x	r_uvsel_ 2x	r_dncov_ flt_byps	r_ycdmx_ rst_sel	r_uv_swa p	r_yc_swa p	r_yrange_ sel	r_rgb_in
Label		R/W	Description						Default
r_dvalid_2x		RW	Wen for 2x clock						0x0
r_uvsel_2x		RW	Uvsel for 2x clock						0x0
r_dncov_flt_byps		RW	LPF bypass when 444 to 422 conversion						0x0
r_ycdmx_rst_sel		RW	YC demux reset selection						0x0
r_uv_swap		RW	For UV swap : 0→UVUV 1→VUVU						0x0
r_yc_swap		RW	For YC swap : 0→UYVY 1→YUYV						0x0
r_yrange_sel		RW							0x0
r_rgb_in		RW	RGB to YUV enable						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x04	DVIN_HSHI	r_hshift_t[7:0]							
0x05	FT	Reserved				r_hshift_t[11:8]			
Label		R/W	Description						Default
r_hshift_t		RW	H pixel shift						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x06	DVIN_VSHI FT	r_vshift_t[7:0]							
0x07		r_vshift_t[15:8]							
0x08		Reserved	r_vshift_t[21:16]						
Label		R/W	Description						Default
r_vshift_t		RW	V pixel shift						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x09	DVIN_TP	r_tp_data_fmt		r_tp_pat_ type	r_tp_bt11 20	r_tp_bt65 6	r_tp_prgs	r_tp_rst	r_tp_en
Label		R/W	Description						Default
r_tp_data_fmt		RW	Test pat fmt;00:8bit YC 01:16bit 422 10:24bit YUV 11:24bit RGB						0x0
r_tp_pat_type		RW	Test pat type ;0: Color bar 1:diagonal line						0x0
r_tp_bt1120		RW	Test pat bt1120 mode ;						0x0
r_tp_bt656		RW	Test pat bt656 mode						0x0
r_tp_prgs		RW	Test pat progressive mode						0x0
r_tp_rst		RW	Reset test pat						0x0
r_tp_en		RW	Test pat enable						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x0a	DVIN_DET	r_sync_s rc_sel	r_hv_det_rsel			r_hv_det _clr	r_hv_det _en	r_sync_d et_en	r_pi_det_ en
Label		R/W	Description						Default

r_sync_src_sel	RW								0x0
r_hv_det_rsel	RW	Hv detection result sel . 0:h total 1:v total 2:h active total 3:h active start 4:v active start other:h total							0x0
r_hv_det_clr	RW	Hv detection clear							0x0
r_hv_det_en	RW	Hv detection enable							0x0
r_sync_det_en	RW	Sync detection enable							0x0
r_pi_det_en	RW	Progressive or interlace detection enable							0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x0b	DVIN_SYN C_STATUS	Reserved			rd_ccir_status			rd_prgs_i ndex	rd_nosyn c
Label		R/W	Description						Default
rd_ccir_status		RO	Crc check result						0x0
rd_prgs_index		RO	Progressive or interlace detection result						0x0
rd_nosync		RO	Sync detection result						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x0c	DVIN_HVD	rd_hvdet_status_t[7:0]							
0x0d	ET_STATU	rd_hvdet_status_t[15:8]							
0x0e	S	rd_hvdet_status_t[23:16]							
Label		R/W	Description						Default
rd_hvdet_status_t		RO	Hv detection result						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x0f	DVIN_TRIG	Reserved							dvin_trig
Label		R/W	Description						Default
dvin_trig		WO	H/v shift trigger						0x0

Video Process and Timing Generation Registers

Base Address : 0x00

Addr.	Name	7	6	5	4	3	2	1	0
0x0e	CORE_VID O_PATH	din_sel	rtm_byps s_656	r_ff_2x_e n	r_ff_en	uv_flip_4 22to444	r_c_deflic ker_en	r_y_deflic ker_en	dvin_dec _en
Label		R/W	Description						Default
din_sel		RW	Input data select.						0x0
rtm_bypss_656		RW	Re_timing module bypass						0x0
r_ff_2x_en		RW	Cross_fifo 2x sampling for 13.5Mhz case						0x0
r_ff_en		RW	Cross_fifo module enable						0x0
uv_flip_422to444		RW	UV flip enable						0x0
r_c_deflicker_en		RW	C signal deflicker enable						0x0
r_y_deflicker_en		RW	Y signal deflicker enable						0x0
dvin_dec_en		RW	Half clock capture for 54MHz clock input						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x0f	CORE_MBI ST	Reserved							dflk_vdly _sel
Label		R/W	Description						Default
dflk_vdly_sel		RW	deflicker vertical delay select . 0: 2 frame delay ; 1: 1frame delay						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x14	CORE_PLL VM_LOCK	Reserved							pllvm_loc k
Label		R/W	Description						Default
pllvm_lock		RW	Pllvm module lock signal						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x15	CORE_PAD _CKO	Reserved						r_dein_s el	r_clk_o_e n
Label		R/W	Description						Default
r_dein_sel		RW	Input data select. 0 : from din_i[29] pin; 1: from am2123a blank_b_i						0x0

r_clk0_en		RW	Clock output enable						0x1
Addr.	Name	7	6	5	4	3	2	1	0
0x30	CORE_RETIM	Reserved					half_line_en	pro_en	fld_inv
0x31		hb_st[7:0]							
0x32							hb_st[10:8]		
0x33		hb_sp[7:0]							
0x34							hb_sp[10:8]		
0x35		vb_odd_st[7:0]							
0x36							vb_odd_st[10:8]		
0x37		vb_odd_sp[7:0]							
0x38							vb_odd_sp[10:8]		
0x39		vb_even_st[7:0]							
0x3a							vb_even_st[10:8]		
0x3b		vb_even_sp[7:0]							
0x3c							vb_even_sp[10:8]		
Label		R/W	Description						Default
vb_even_sp		RW	Retiming module vblank signal even field stop line						0x0
vb_even_st		RW	Retiming module vblank signal even field start line						0x0
vb_odd_sp		RW	Retiming module vblank signal odd field stop line						0x0
vb_odd_st		RW	Retiming module vblank signal odd field start line						0x0
hb_sp		RW	Retiming module hblank signal stop pixel						0x0
hb_st		RW	Retiming module hblank signal start pixel						0x0
half_line_en		RW	Vblank signal insert in middle of a line enable						0x0
pro_en		RW	Progressive video enable						0x0
fld_inv		RW	field inv enable						0x0

TV Encode Registers

Base Address : 0x60

Addr.	Name	7	6	5	4	3	2	1	0
0x00	tve_mode	Reserved	tve_tv_type	tve_pal60	Reserved	menu_mode_en	Reserved	tve_svdo_out_en	tve_cvbs_out_en
Label		R/W	Description						Default
tve_tv_type		RW	TV type selection. 0: NTSC; 1: PAL-M; 2: PAL-Combi; 3: PAL						0x0
tve_pal60		RW	PAL 60Hz mode enable, active high						0x0
menu_mode_en		RW	The manul to moudulate the tv encode wavefrom enable signal.						0x0
tve_svdo_out_en		RW	S-VIDEO output enable, active high						0x0
tve_cvbs_out_en		RW	CVBS output enable, active high						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x01	tve_shift	Reserved				tve_flip_in_field	Reserved	tve_line_shift[1:0]	
Label		R/W	Description						Default
tve_flip_in_field		RW	TVE input field flip						0x0
tve_line_shift[1:0]		RW	line shift 0,1: not shift; 2:up shift 1 line; 3: down shift 1 line						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x02	tve_p2i	Reserved				tve_p2i_fld_flip	tve_uv444to422flt_bypass	tve_p2i_line_inv	tve_p2i_en
Label		R/W	Description						Default
tve_p2i_fld_flip		RW	field index flip after progressive to interlace convert						0x0
tve_uv444to422flt_byps		RW	UV 444 to 422 conversion UV decimation filter bypass						0x0

tve_p2i_line_inv	RW	line index invert in progressive to interlace convert	0x0
tve_p2i_en	RW	progressive to interlace convert enable	0x1

Addr.	Name	7	6	5	4	3	2	1	0
0x03	tve_misc	Reserved		tve_color_bar_type	tve_color_bar_en	tve_setup_7p5ire	tve_blacker	tve_slew_off	tve_full_width
Label		R/W	Description						Default
tve_color_bar_type		RW	built in color bar type selection						0x0
tve_color_bar_en		RW	built in color bar enable, active high						0x0
tve_setup_7p5ire		RW	7.5IRE setup in NTSC mode						0x0
tve_blacker		RW	blacker than blank enable. Active high						0x0
tve_slew_off		RW	transition slew disable. 1: disable; 0: enable						0x0
tve_full_width		RW	1: full width (720pixels) in horizontal ; 0: non-full-width(704pixels) in horizontal						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x04	tve_y_act	Reserved		tve_y_act_bgn[5:0]					
Label		R/W	Description						Default
tve_y_act_bgn[5:0]		RW	YUV active data position in P2I, change the value from 0 to 63, the picture position shift from left to right						0x20

Addr.	Name	7	6	5	4	3	2	1	0
0x06	tve_intp	Reserved					tve_2nd_intp_by ps	tve_uv_1st_intp_byps	tve_y_1st_intp_byps
Label		R/W	Description						Default
tve_2nd_intp_byps		RW	2nd stage 2x interpolation disable. 1: disable; 0: enable						0x0
tve_uv_1st_intp_byps		RW	C 1st stage 2x interpolation disable. 1: disable; 0: enable						0x0
tve_y_1st_intp_byps		RW	Y 1st stage 2x interpolation disable. 1: disable; 0: enable						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x07	tve_lpf	tve_c_lpf_sel[7:6]	Reserved	tve_c_lpf_byps	tve_y_lpf_sel[3:2]	Reserved			tve_y_lpf_byps
Label		R/W	Description						Default
tve_c_lpf_sel[7:6]		RW	C low pass filter BW selection. From 0 to 3, filter BW from high to low						0x1
tve_c_lpf_byps		RW	C low pass filter disable. 1: disable; 0: enable						0x0
tve_y_lpf_sel[3:2]		RW	Y low pass filter BW selection. From 0 to 3, filter BW from high to low						0x0
tve_y_lpf_byps		RW	Y low pass filter disable. 1: disable; 0: enable						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x08	tve_gamma0	tve_gamma0[7:0]							
Label		R/W	Description						Default
tve_gamma0[7:0]		RW	piece-wise linear gamma point 0 value						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x09	tve_gamma1	tve_gamma1[7:0]							
Label		R/W	Description						Default
tve_gamma1[7:0]		RW	piece-wise linear gamma point 1 value						0x20

Addr.	Name	7	6	5	4	3	2	1	0
0x0a	tve_gamma2	tve_gamma2[7:0]							
Label		R/W	Description						Default
tve_gamma2[7:0]		RW	piece-wise linear gamma point 2 value						0x40

Addr.	Name	7	6	5	4	3	2	1	0
0x0b	tve_gamma3	tve_gamma3[7:0]							
Label		R/W	Description						Default
tve_gamma3[7:0]		RW	piece-wise linear gamma point 3 value						0x60

Addr.	Name	7	6	5	4	3	2	1	0
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0x0c	tve_gamma4	tve_gamma4[7:0]							
Label		R/W	Description						Default
tve_gamma4[7:0]		RW	piece-wise linear gamma point 4 value						0x80
Addr.	Name	7	6	5	4	3	2	1	0
0x0d	tve_gamma5	tve_gamma5[7:0]							
Label		R/W	Description						Default
tve_gamma5[7:0]		RW	piece-wise linear gamma point 5 value						0xa0
Addr.	Name	7	6	5	4	3	2	1	0
0x0e	tve_gamma6	tve_gamma6[7:0]							
Label		R/W	Description						Default
tve_gamma6[7:0]		RW	piece-wise linear gamma point 6 value						0xc0
Addr.	Name	7	6	5	4	3	2	1	0
0x0f	tve_gamma7	tve_gamma7[7:0]							
Label		R/W	Description						Default
tve_gamma7[7:0]		RW	piece-wise linear gamma point 7 value						0xe0
Addr.	Name	7	6	5	4	3	2	1	0
0x10	tve_gamma8	tve_gamma8[7:0]							
Label		R/W	Description						Default
tve_gamma8[7:0]		RW	piece-wise linear gamma point 8 value						0xff
Addr.	Name	7	6	5	4	3	2	1	0
0x11	tve_gain_en	Reserved						tve_uv_gain_en	tve_y_gain_en
Label		R/W	Description						Default
tve_uv_gain_en		RW	U gain control enable, active high						0x1
tve_y_gain_en		RW	Y gain control enable, active high						0x1
Addr.	Name	7	6	5	4	3	2	1	0
0x12	tve_ygain	tve_y_gain[7:0]							
Label		R/W	Description						Default
tve_y_gain[7:0]		RW	Y gain value, range: (0~2)x128						0x4b
Addr.	Name	7	6	5	4	3	2	1	0
0x13	tve_ugain1	tve_u_gain1[7:0]							
Label		R/W	Description						Default
tve_u_gain1[7:0]		RW	U gain in 1st quadrant, range: (0~2)x128						0x40
Addr.	Name	7	6	5	4	3	2	1	0
0x14	tve_ugain2	tve_u_gain2[7:0]							
Label		R/W	Description						Default
tve_u_gain2[7:0]		RW	U gain in 2nd quadrant, range: (0~2)x128						0x40
Addr.	Name	7	6	5	4	3	2	1	0
0x15	tve_ugain3	tve_u_gain3[7:0]							
Label		R/W	Description						Default
tve_u_gain3[7:0]		RW	U gain in 3rd quadrant, range: (0~2)x128						0x40
Addr.	Name	7	6	5	4	3	2	1	0
0x16	tve_ugain4	tve_u_gain4[7:0]							
Label		R/W	Description						Default
tve_u_gain4[7:0]		RW	U gain in 4th quadrant, range: (0~2)x128						0x40
Addr.	Name	7	6	5	4	3	2	1	0
0x17	tve_vgain1	tve_v_gain1[7:0]							
Label		R/W	Description						Default
tve_v_gain1[7:0]		RW	V gain in 1st quadrant, range: (0~2)x128						0x5b

Addr.	Name	7	6	5	4	3	2	1	0
0x18	tve_vgain2	tve_v_gain2[7:0]							
Label		R/W	Description						Default
tve_v_gain2[7:0]		RW	V gain in 2nd quadrant, range: (0~2)x128						0x5b

Addr.	Name	7	6	5	4	3	2	1	0
0x19	tve_vgain3	tve_v_gain3[7:0]							
Label		R/W	Description						Default
tve_v_gain3[7:0]		RW	V gain in 3rd quadrant, range: (0~2)x128						0x5b

Addr.	Name	7	6	5	4	3	2	1	0
0x1a	tve_vgain4	tve_v_gain4[7:0]							
Label		R/W	Description						Default
tve_v_gain4[7:0]		RW	V gain in 4th quadrant, range: (0~2)x128						0x5b

Addr.	Name	7	6	5	4	3	2	1	0
0x1b	tve_burst_en	Reserved						tve_burst_on_u	tve_burst_on_v
Label		R/W	Description						Default
tve_burst_on_u		RW	color burst on u enable						0x1
tve_burst_on_v		RW	color burst on v enable						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x1c	tve_burst_st	tve_burst_st[7:0]							
Label		R/W	Description						Default
tve_burst_st[7:0]		RW	color burst start position						0x8c

Addr.	Name	7	6	5	4	3	2	1	0
0x1d	tve_burst_sp	tve_burst_sp[7:0]							
Label		R/W	Description						Default
tve_burst_sp[7:0]		RW	color burst stop position						0xd4

Addr.	Name	7	6	5	4	3	2	1	0
0x1e	tve_burst_level	tve_burst_level[7:0]							
Label		R/W	Description						Default
tve_burst_level[7:0]		RW	color burst level						0x72

Addr.	Name	7	6	5	4	3	2	1	0
0x1f	tve_upq_init	tve_upq_init[7:0]							
Label		R/W	Description						Default
tve_upq_init[7:0]		RW	color burst initial phase						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x20	tve_sync_base_lv	tve_sync_base_lv[7:0]							
Label		R/W	Description						Default
tve_sync_base_lv[7:0]		RW	Sync base level						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x21	tve_bright_lv	tve_bright_lv[7:0]							
Label		R/W	Description						Default
tve_bright_lv[7:0]		RW	Picture brithtness level, the larger the value, more bright on picture						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x22	tve_blank_lv	Reserved				tve_blank_lv[3:0]			
Label		R/W	Description						Default
tve_blank_lv[3:0]		RW	Blank level, valid in blacker than blank						0x4

Addr.	Name	7	6	5	4	3	2	1	0
0x23	tve_bot_thrd	tve_bot_thrd[7:0]							
Label		R/W	Description						Default
tve_bot_thrd[7:0]		RW	bottom threshold of Y						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x24	tve_top_thrd	tve_top_thrd[7:0]							
Label		R/W	Description						Default
tve_top_thrd[7:0]		RW	top threshold of Y						0xff

Addr.	Name	7	6	5	4	3	2	1	0
0x25	tve_svdo_c hro_gain	tve_svdo_chro_gain[7:0]							
Label		R/W	Description						Default
tve_svdo_chro_gain		RW	S-VIDEO chroma gain, range: (0~2)*128						0xce

Addr.	Name	7	6	5	4	3	2	1	0
0x26	tve_svdo_c hro_ofst	tve_svdo_chro_ofst[7:0]							
Label		R/W	Description						Default
tve_svdo_chro_ofst[7:0]		RW	S-VIDEO chroma offset, range: -128~127						0xb2

Addr.	Name	7	6	5	4	3	2	1	0
0x27	tve_itlc_ln_s hift_l	tve_itlc_ln_shift_l[7:0]							
0x28	tve_itlc_ln_s hift_h	Reserved						tve_itlc_ln_shift_h[9:8]	
Label		R/W	Description						Default
tve_itlc_ln_shift[9:0]		RW	Control the line counter when interlace input						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x29	tve_itlc_pxl_ shift_l	tve_itlc_pxl_shift_l[7:0]							
0x2a	tve_itlc_pxl_ shift_h	Reserved			tve_itlc_p xl_shift_d is	Reserved	tve_itlc_pxl_shift_h[10:8]		
Label		R/W	Description						Default
tve_itlc_pxl_shift[10:0]		RW	Control the pixel counter when interlace input						0x693
tve_itlc_pxl_shift_dis		RW	Disable the pixel counter when interlace input just fot test						0x0

DACMUX and Hot Plug Registers

Base Address: 0x90

Addr.	Name	7	6	5	4	3	2	1	0
0x00	hp_enable	Reserved			flip_fldin	Manu_m ode_en	ypbpr_en	cvbs_en	hp_en
Label		R/W	Description						Default
flip_fldin		RW	HPD input field flip						0x0
manu_mode_en		RW	Manual mode enable						0x0
ypbpr_en		RW	Output video mode select 1:YPBPR 0:CVBS/VGA						0x0
cvbs_en		RW	Output video mode select 1:CVBS 0:YPBPR/VGA						0x1
hp_en		RW	Hotplug function enable						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x01	hp_dtc_bgn_ _l	hp_dtc_bgn[7:0]							
0x02	hp_dtc_bgn_ _h	Reserved					hp_dtc_bgn[10:8]		
Label		R/W	Description						Default

hp_dtc_bgn[10:0]		RW	HP detect begin line by line counter						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x03	hp_dtc_end_l	hp_dtc_end[7:0]							
0x04	hp_dtc_end_h	Reserved					hp_dtc_wth[10:8]		
Label		R/W	Description						Default
hp_dtc_end[10:0]		RW	HP detect end line by line counter						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x05	dig_sm_line_l	dig_sm_line[7:0]							
0x06	dig_sm_line_h	Reserved					dig_sm_line[10:8]		
Label		R/W	Description						Default
dig_sm_line[10:0]		RW	HP detect sampling line						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x07	fst_sm_bgn_l	fst_sm_bgn[7:0]							
0x08	fst_sm_bgn_h	Reserved				fst_sm_bgn[11:8]			
Label		R/W	Description						Default
fst_sm_bgn[11:0]		RW	HP detect fist sampling						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x09	sec_sm_bg_n_l	sec_sm_bgn[7:0]							
0x0a	sec_sm_bg_n_h	Reserved				sec_sm_bgn[11:8]			
Label		R/W	Description						Default
sec_sm_bgn[11:0]		RW	HP detect second sampling						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x0b	squ_line_bgn_n_l	squ_line_bgn [7:0]							
0x0c	squ_line_bgn_n_h	Reserved					squ_line_bgn [10:8]		
Label		R/W	Description						Default
squ_line_bgn [10:0]		RW	The vertical begin position when square wave stack on VGA signal						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x0d	squ_line_end_d_l	squ_line_end [7:0]							
0x0e	squ_line_end_d_h	Reserved					squ_line_end [10:8]		
Label		R/W	Description						Default
squ_line_end [10:0]		RW	The vertical end position when square wave stack on VGA signal						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x0f	squ_h_bgn_l	squ_h_bgn [7:0]							
0x10	squ_h_bgn_h	Reserved				squ_h_bgn [11:8]			
Label		R/W	Description						Default
squ_h_bgn [11:0]		RW	The horizontal begin position when square wave stack on VGA signal						0x0
Addr.	Name	7	6	5	4	3	2	1	0
0x11	squ_h_end_l	squ_h_end [7:0]							

0x12	squ_h_end_h	Reserved				squ_h_end [11:8]			
Label		R/W	Description						Default
squ_h_end [11:0]		RW	The horizontal end position when square wave stack on VGA signal						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x13	hp_dig_stat_us_rd	Reserved				dig_ana_en	hp_r_stat_e	hp_g_sta_te	hp_b_sta_te
Label		R/W	Description						Default
dig_ana_en		RO	For test						---
hp_r_state		RO	Hotplug detect R channel state						---
hp_g_state		RO	Hotplug detect G channel state						---
hp_b_state		RO	Hotplug detect B channel state						---

Addr.	Name	7	6	5	4	3	2	1	0
0x14	hp_ana_status	reg_ana_en	sw_en	Reserved			ana_hp_b	ana_hp_g	ana_hp_r
Label		R/W	Description						Default
reg_ana_en		RW	Register to analog hpd enable for test						0x0
sw_en		RW	sw conterl enable for test						0x0
ana_hp_b		RO	Analog module hpd state for test						---
ana_hp_g		RO	Analog module hpd state for test						---
ana_hp_r		RO	Analog module hpd state for test						---

Addr.	Name	7	6	5	4	3	2	1	0
0x15	hp_vga_squ	vga_squ[7:0]							
Label		R/W	Description						Default
vga_squ[7:0]		RW	Square wave on vga single						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x16	hp_byps	Reserved					hp_v_by_ps	hp_u_by_ps	hp_y_by_ps
Label		R/W	Description						Default
hp_v_byps		RW	Hpd interpolation on y channel byps						0x0
hp_u_byps		RW	Hpd interpolation on u channel byps						0x0
hp_y_byps		RW	Hpd interpolation on v channel byps						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x17	reg_to_dac_l	reg_to_dac[7:0]							
0x18	reg_to_dac_h	Reserved						reg_to_dac[9:8]	
Label		R/W	Description						Default
reg_to_dac[9:0]		RW	The value form register to vdac for test						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x19	data_sl_t	ubout_ch_sl_t	ygout_ch_sl_t		t_ch_sl_t	r_ch_sl_t	b_ch_sl_t	g_ch_sl_t	
Label		R/W	Description						Default
ubout_ch_sl_t		RW	UB channel output select 00: B 01:G 10:R 11:T						0x0
ygout_ch_sl_t		RW	Yg channel output select 00: G 01:B 10:R 11:T						0x0
t_ch_sl_t		RW	T channel data 0:data from register reg_to_dac[9:0]						0x0
r_ch_sl_t		RW	R channel data 0:data form compout VGA/YPBPR v/r 1:form tve SVIDEO C						0x0
b_ch_sl_t		RW	B channel data 0:data form compout VGA/YPBPR u/b 1:form tve SVIDEO Y						0x0
g_ch_sl_t		RW	G channel data 0:form compout VGA/YPBPR y/g 1:form tve AV						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x1a	Timing_sl_t	Reserved					timout_sl_t	vrout_ch_sl_t	
Label		R/W	Description						Default
timout_sl_t		RW	Timing out select 0:vga syc output						0x0
vrout_ch_sl_t		RW	VR channel output select 00: R 01:B 10:G 11:T						0x0

PLLVM Registers

Base Address: 0x20

Addr.	Name	7	6	5	4	3	2	1	0
0x0	PLLVM_PWR	pllvm_icps[3:0]				pllvm_bgre f_start	pllvm_bgre f_pwd	pllvm_reg3 318_pwd	pllvm_pwd b
Label		R/W	Description						Default
pllvm_icps[3:0]		RW	VCP work current select 0000: 0 uA 0001: 5 uA 0010: 10uA 0100: 20uA 1000: 40 uA 1111: 75uA						0x5
pllvm_bgref_start		RW	Start BGREF circuit 0: start circuit work normal. 1: Start BGREF (need up rise trigger , BGREF can start up);						0x0
pllvm_bgref_pwd		RW	BGREF Power down 0: Power on 1: Power down						0x1
pllvm_reg3318_pwd		RW	REG3318 Power down 0: Power on 1: Power down						0x1
pllvm_pwdb		RW	Power down PLL 0: Power Down 1: Power on						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x1	PLLVM_N	pllvm_n[7:0]							
Label		R/W	Description						Default
pllvm_n[7:0]		RW	8 bits Input clock divider, out is CKREF						0x7

Addr.	Name	7	6	5	4	3	2	1	0
0x2	PLLVM_M_L	pllvm_m[7:0]							
0x3	PLLVM_M_H	Reserved				pllvm_m[11:8]			
Label		R/W	Description						Default
pllvm_m[11:0]		RW	12 bits PLLVM feedback divider						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x4	PLLVM_FN_L	pllvm_fn[7:0]							
0x5	PLLVM_FN_M	pllvm_fn[15:8]							
0x6	PLLVM_FN_H	Reserved			pllvm_fnen	pllvm_fn[19:16]			
Label		R/W	Description						Default
pllvm_fn[19:0]		RW	20 bits PLLVM fraction divider						0x0
pllvm_fnen		RW	Enable to fraction 0: Turn off fraction 1: Turn on fraction						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x7	PLLVM_MD	pllvm_md[7:0]							
0x8	PLLVM_VD	pllvm_vd[7:0]							
Label		R/W	Description						Default
pllvm_md[7:0]		RW	MCLK out POST divider; posd8[1:0],00:div1, 01:div2, 10:div4 11:div8; posd8[2],0:div3 disable,1:div3 enable posd8[3],0:div5, 1:div7 posd8[4],0:div9 disable,1:div9 posd8[6:5]:00:select div1248,01:select div3, 10: select div5/7,11:select div9 posd8[7] 0:post divider bypass,1:post divider by 2.						0x0
pllvm_vd[7:0]		RW	VCLK out POST divider; posd8[1:0],00:div1, 01:div2, 10:div4 11:div8; posd8[2],0:div3 disable,1:div3 enable posd8[3],0:div5, 1:div7 posd8[4],0:div9 disable,1:div9 posd8[6:5]:00:select div1248,01:select div3, 10: select div5/7,11:select div9 posd8[7] 0: post divider bypass,1:post divider by 2.						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0x9	PLLVM_DIVM UX	pllvm_vco_ fs	Reserved	pllvm_vd_k s	pllvm_vd_div4s[1:0]		pllvm_vd_div2s[1:0]		pllvm_vd_v 4cksxtal
Label		R/W	Description						Default
PLLVM_VCO_FS		RW	VCO band select						0x1

		0: VCO run at low gain(600M--900M) 1: VCO run at high gain(900M--1200M)	
pllvm_vd_ks	RW	PLLV post divider select KS=0,bypass KS=1,div2;	0x0
pllvm_vd_div4s[1:0]	RW	NS_VCLK divider, 00:bypass; 01:div2; 10:div4; 11:NS_XTAL;	0x0
pllvm_vd_div2s[1:0]	RW	NS_V2CLK divder select, 00:div1, 01:div2, 10:select XTAL, 11:vss18	0x0
pllvm_vd_v4cksxtal	RW	NS_V4CLK select 0: V4CLK, 1:NS_XTAL;	0x0

Addr.	Name	7	6	5	4	3	2	1	0
0xa	PLLVM_CKIS EL	pllvm_rsel[1:0]		Reserved		pllvm_fnem icks	pllvm_ckis	pllvm_cki_i nvs	pllvm_cki_ div2s
Label		R/W	Description						Default
pllvm_rsel[1:0]		RW	Adjust LDO output voltage 00 : 1.893---2.1v 2V 01 : 1.74---1.89 v 1.79V 10 : 1.83---2v 1.9V 11 : 2.01---2.23 v 2.1V						0x2
pllvm_fnemicks		RW	Select clock to fraction 0: CKREF 1: Feedback clock CKFB						0x0
pllvm_ckis		RW	Input clock select 0: CKOSC 1: CLKIN						0x1
pllvm_cki_invs		RW	Input clock invert select 0: Normal input clock 1: Invert of input clock						0x0
pllvm_cki_div2s		RW	Input clock div2 enable 0: Input clock bypass 1: CLKIN div2						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0xb	PLLVM_LOCK	pllvm_test_en	pllvm_test_s[1:0]		pllvm_lock_s[1:0]		pllvm_lock_r[1:0]		pllvm_lock_en
Label		R/W	Description						Default
pllvm_test_en		RW	CKTEST test enable 0: low voltage 1: open test path						0x0
pllvm_test_s[1:0]		RW	CKTEST path select 00: CKFB , 01: V4CLK(not OSC), 10=11 ;CKREF						0x0
pllvm_lock_s[1:0]		RW	Skew control for lock indicator;						0x0
pllvm_lock_r[1:0]		RW	Skew control for lock indicator;						0x0
pllvm_lock_en		RW	Enable for lock detect circuit 0: Disable lock circuit ,1: Enable lock circuit						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0xc	PLLVM_OCLK EN	Reserved		pllvm_ssce n	pllvm_sms	pllvm_vco_ rst	pllvm_v18_ testen	pllvm_audc k_en	pllvm_mem ck_byps
Label		R/W	Description						Default
pllvm_ssцен		RW	Frequency Modulator 0: Disable Frequency Modulator 1: Enable Frequency Modulator						0x0
pllvm_sms		RW	Break up concentrated energy to prevent EMI 0: Disable 1: Enable						0x0
pllvm_vco_rst		RW	VCO reset 0: Normal work 1: Reset VCO						0x0
pllvm_v18_testen		RW	REG3318 out voltage test enable 0:low voltage 1: REG3318 out voltage can test AUDIO clock enable 0: low voltage 1: OSC/2						0x0
pllvm_audck_en		RW	AUDIO clock enable 0: low voltage 1: OSC/2						0x0
pllvm_memck_byps		RW	MCLK bypass 0: Normal path 1: Bypass NS_CKOSC to NS_MCLK						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0xd	PLLVM_SP	Reserved	pllvm_mf[3:0]				pllvm_ma[2:0]		
Label		R/W	Description						Default
pllvm_mf[3:0]		RW	Spread Spectrum clock modulation frequency selector						0x0
pllvm_ma[2:0]		RW	Spread Spectrum clock modulation current selector						0x0

Addr.	Name	7	6	5	4	3	2	1	0
0xe	PLLVM_TRIG	Reserved							pllvm_trigger
Label		R/W	Description						Default
pllvm_trigger		RW	PLLVM Trigger bit.						0x0

VDAC Registers

Base Address: 0xAB

Addr.	Name	7	6	5	4	3	2	1	0
0x0	REG_PDZ	REG_B1_PDZ	REG_B2_PDZ	RG_CK_INV	BPDZ	GPDZ	RPDZ	RG_CK_PD	BGPDZ
Label		R/W	Description						Default
REG_B1_PDZ		RW	0: IOG/R/B (mA) = 11,355 × 1.2 (V)/RSET (Ω) (1) 1: IOG,IOB ,IOR(mA) = 7922.5 × 1.2 (V)/RSET (Ω)						0
REG_B2_PDZ		RW	Together with the register"REG_B1_PDZ"to keep the output voltage constant ,the external resistor normal work:0						0
RG_CK_INV		RW	Inverse Clock 0: normal work 1: Inverse						0
BPDZ		RW	B channel power down 1: normal work 0: power down						0
GPDZ		RW	G channel power down 1: normal work 0: power down						0
RPDZ		RW	R channel power down 1: normal work 0: power down						0
RG_CK_PD		RW	Clock power down 1: power down 0: normal work						1
BGPDZ		RW	BG&V2I power down 1: normal work 0: power down						0

Addr.	Name	7	6	5	4	3	2	1	0
0x1	REG_RGBEN	Reserved		B1EN	B0ENZ	G1EN	G0ENZ	R1EN	R0ENZ
Label		R/W	Description						Default
B1EN		RW	B channel set 0: normal work 1: set						0
B0ENZ		RW	B channel reset 0: reset 1: normal work						1
G1EN		RW	G channel set 0: normal work 1: set						0
G0ENZ		RW	G channel reset 0: reset 1: normal work						1
R1EN		RW	R channel set 0: normal work 1: set						0
R0ENZ		RW	R channel reset 0: reset 1: normal work						1

Addr.	Name	7	6	5	4	3	2	1	0
0x2	REG_HPDEN	RG_HPD_VTHSEL[3:0]				RG_VDAC_PLL_VEN	Reserved	ABISTEN	RG_HPD_EN
Label		R/W	Description						Default
RG_HPD_VTHSEL		RW	Hot plug detect compare voltage						1000
RG_VDACPLL_VEN		RW	VDAC VBG12 output to PLLVM enable 0: disable 1: enable						1
ABISTEN		RW	VDAC output to ADC input enable 0: disable 1: enable						0
RG_HPD_EN		RW	Hot plug detect enable 0: disable 1: enable						0